

JOHANNESBURG BOTANICAL, South Africa

WMO: 683610

Lat: 26.1565S

Lon: 27.9988E

Elev: 5328

StdP: 12.08

Time Zone: 2.00 (E02)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.7	35.8	8.3	10.2	60.3	12.0	12.3	58.1	9.3	52.5	7.8	54.3	1.7	220	0.324

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.6	89.4	59.7	87.1	59.4	84.6	59.5	67.5	76.9	66.2	76.0	65.1	75.4	3.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	112.3	70.3	63.0	105.2	68.7	62.4	103.0	68.3	35.4	77.7	34.2	76.5	33.2	75.0	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	7.2	6.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			26.3	68.4	2.0	0.9	24.9	69.1	23.8	69.6	22.6	70.1	21.2	70.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.1	70.7	69.9	68.0	62.4	57.4	52.8	52.2	56.7	63.5	66.9	67.5	70.2		
	DBStd	7.91	3.59	3.04	3.72	4.32	4.68	3.87	3.77	5.55	5.75	5.31	5.04	4.19		
	HDD50	59	0	0	0	0	8	18	21	11	1	0	0	0		
	HDD65	1548	3	3	15	99	235	367	396	259	91	42	30	8		
	CDD50	4856	640	556	557	372	239	101	90	220	407	523	526	625		
	CDD65	872	179	139	108	21	1	0	0	3	47	99	107	168		
	CDH74	8264	1234	945	856	264	79	4	4	173	917	1332	1101	1355		
	CDH80	2220	357	200	168	34	3	0	0	10	238	454	337	419		
(14) Wind	WSAvg	2.0	2.1	1.9	1.7	1.6	1.6	1.8	1.7	2.1	2.2	2.4	2.4	2.3		
(15) Precipitation	PrecAvg	28.1	4.9	4.5	3.9	1.5	0.7	0.2	0.0	0.2	0.6	2.6	3.7	4.7		
	PrecMax	43.2	8.9	12.9	13.0	4.9	4.0	0.7	0.4	1.3	2.3	5.8	8.2	8.2		
	PrecMin	20.1	1.5	0.7	0.7	0.0	0.0	0.0	0.0	0.0	0.0	0.8	0.4	1.5		
	PrecStd	6.1	1.9	3.1	2.8	1.3	1.0	0.2	0.1	0.3	0.6	1.4	1.8	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.3	87.9	86.4	84.0	80.0	73.7	73.8	81.0	88.0	91.5	91.3	91.7		
		MCWB	61.6	61.5	60.5	59.8	54.3	50.0	48.3	53.1	56.1	58.1	57.6	61.5		
	2%	DB	87.4	84.7	84.3	80.4	75.7	71.4	71.3	78.7	85.7	88.0	87.5	87.9		
		MCWB	62.1	64.1	60.2	58.6	52.3	49.3	48.0	52.6	54.7	56.9	58.5	61.7		
	5%	DB	84.2	82.5	82.0	77.0	73.5	68.4	68.3	75.4	82.7	85.1	84.0	84.4		
		MCWB	62.5	63.2	60.2	56.3	52.2	47.9	47.6	50.7	53.9	56.8	58.7	61.6		
	10%	DB	80.8	80.2	79.0	74.3	70.8	66.1	66.1	71.9	80.3	82.1	80.5	82.0		
		MCWB	62.9	63.0	60.4	56.2	52.1	47.5	47.4	49.7	53.1	56.6	59.8	61.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.9	78.6	66.8	62.9	59.0	54.0	60.4	57.5	59.1	62.7	73.2	67.5		
		MCDB	76.6	79.7	78.0	73.6	68.8	65.7	61.7	71.9	78.3	78.8	74.4	79.2		
	2%	WB	67.1	67.6	64.8	61.9	56.5	51.9	50.7	54.9	57.6	61.4	64.4	65.7		
		MCDB	76.6	76.6	75.0	72.5	69.0	65.4	64.2	72.3	76.8	77.3	77.5	77.6		
	5%	WB	65.9	66.1	63.6	60.7	55.1	50.3	49.1	53.1	56.4	60.2	62.8	64.7		
		MCDB	75.3	75.4	73.8	71.3	67.5	63.7	63.5	70.8	75.9	75.8	76.5	76.9		
	10%	WB	65.0	64.9	62.5	59.3	53.6	49.0	47.8	51.5	55.2	59.0	61.6	63.8		
		MCDB	74.5	74.7	72.7	69.3	66.2	62.3	63.1	68.5	74.7	74.4	74.6	76.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.6	20.2	21.1	22.8	25.3	26.6	27.4	28.3	28.9	26.8	23.6	21.3		
		MCDBR	25.6	24.6	25.9	27.0	28.8	30.1	30.7	31.7	32.6	31.4	28.7	26.2		
		MCWBR	8.6	10.5	9.1	10.4	11.9	13.4	14.1	13.5	12.5	10.9	9.9	9.0		
	5% WB	MCDBR	18.6	19.5	19.9	20.4	24.1	25.5	26.9	27.7	28.5	25.4	23.4	21.7		
		MCWBR	7.9	10.0	8.4	9.1	11.5	12.9	14.2	12.6	11.5	10.4	10.2	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.386	0.389	0.376	0.328	0.292	0.274	0.276	0.329	0.370	0.403	0.363	0.371		
	taud		2.324	2.316	2.331	2.438	2.518	2.540	2.522	2.365	2.265	2.206	2.358	2.355		
	Ebn at Noon		304	298	292	292	291	291	294	287	289	290	308	309		
	Edn at Noon		44	43	40	33	27	25	27	35	42	48	42	43		
(44) All-Sky Solar Radiation	RadAvg		2103	1984	1760	1508	1377	1250	1357	1602	1912	2047	2141	2160		
	RadStd		225	178	160	133	70	60	36	75	87	142	191	130		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+1.19	+1.35	N/A	N/A	N/A	N/A	N/A	-326	+421	+157

Nomenclature: See separate page